

# Seasonal ARIMA models

FISH 550 – Applied Time Series Analysis [Download Rmd pdf](#)

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10 Apr 2025

# Seasonality

Load the chinook salmon data set

```
load("chinook.RData")  
head(chinook)
```

| ##   | Year | Month | Species | log.metric.tons | metric.tons |
|------|------|-------|---------|-----------------|-------------|
| ## 1 | 1990 | Jan   | Chinook | 3.397858        | 29.9        |
| ## 2 | 1990 | Feb   | Chinook | 3.808882        | 45.1        |
| ## 3 | 1990 | Mar   | Chinook | 3.511545        | 33.5        |
| ## 4 | 1990 | Apr   | Chinook | 4.248495        | 70.0        |
| ## 5 | 1990 | May   | Chinook | 5.200705        | 181.4       |
| ## 6 | 1990 | Jun   | Chinook | 4.371976        | 79.2        |

The data are monthly and start in January 1990. To make this into a ts object do

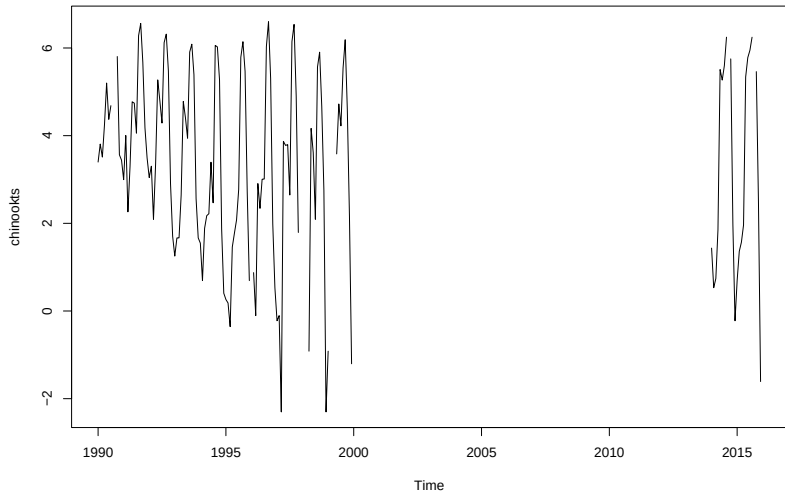
```
chinookts <- ts(chinook$log.metric.tons, start=c(1990,1),  
               frequency=12)
```

start is the year and month and frequency is the number of months in the year.

Use ?ts to see more examples of how to set up ts objects.

## Plot seasonal data

```
plot(chinookts)
```



# Seasonal ARIMA model SARIMA

Seasonally differenced data, e.g. chinook data January 1990 - chinook data January 1989.

$$Z_t = x_t - x_{t+s} - m$$

Basic structure of a seasonal AR model

seasonal differenced data = AR(seasonal) + AR(non-seasonal) + AR(p+season)

For example, SARIMA model could capture this process

1. AR(seasonal) January in  $t$  is correlated with January in  $t - 1$
2. AR(non-seasonal) January differences are correlated with February differences
3. AR( $p$ +season) appears because of 1 and 2

$$z_t = \text{AR}(1) + \text{AR}(12) + \text{AR}(1+12)$$

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \phi_1 \Phi_1 z_{t-13}$$

# Notation

ARIMA (p,d,q)(ps,ds,qs)S

ARIMA (non-seasonal part)(seasonal part)Frequency

ARIMA (non-seasonal) means  $t$  correlated with  $t - 1$

ARIMA (seasonal) means  $t$  correlated with  $t - s$



## Examples

ARIMA (1,0,0)(1,0,0)[12]

What data are we modeling? Get the differences

$$z_t = x_t - m$$

Write out the AR parts with  $z_t$

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \phi_1 \Phi_1 z_{t-13} + w_t$$

Write out the MA parts, the  $w_t$ . No MA in this model.

$$w_t = e_t$$

ARIMA (1,0,0)(1,1,0)[12]

Figure out  $z_t$ . Just a seasonal difference.

$$z_t = x_t - x_{t-12} - m$$

Write out the AR parts with  $z_t$

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \phi_1 \Phi_1 z_{t-13} + w_t$$

Write out the MA parts, the  $w_t$ . No MA in this model.  $w_t$  is white noise.

$$w_t = e_t$$

## Seasonal random walk model

ARIMA(0,0,0)(0,1,0)[12]

expected January 1990 = January 1989 + constant mean

Figure out  $z_t$ .  $m$  is the mean seasonal difference.

$$z_t = x_t - x_{t-12} - m$$

Write out the AR parts with  $z_t$ . No AR part.

$$z_t = w_t$$

Write out the MA parts, the  $w_t$ .

$$w_t = e_t$$

## Seasonal random walk model with random trend

ARIMA(0,1,0)(0,1,0)[12]

expected Feb 1990 = Feb 1989 + (Jan 1990 - Jan 1989)

Figure out  $z_t$ .  $m$  is the mean seasonal difference.

$$z_t = (x_t - x_{t-12}) - (x_{t-1} - x_{t-13}) - m$$

Write out the AR parts with  $z_t$ . No AR part.

$$z_t = w_t$$

Write out the MA parts, the  $w_t$ .

$$w_t = e_t$$

## airline model

ARIMA(0, 1, 1)(0, 1, 1)[12]

Figure out  $z_t$ .

$$z_t = (x_t - x_{t-12}) - (x_{t-1} - x_{t-13}) - m$$

Write out the AR parts with  $z_t$ . No AR part.

$$z_t = w_t$$

Write out the MA parts, the  $w_t$ .

$$w_t = e_t - \theta_1 e_{t-1} - \Theta_1 e_{t-12} + \theta_1 \Theta_1 e_{t-13}$$

## Example with longer lags

ARIMA (2,0,1)(1,0,2)[12]

What data are we modeling? Get the differences

$$z_t = x_t - m$$

Write out the AR parts with  $z_t$

$$z_t = \phi_1 z_{t-1} + \phi_2 z_{t-2} + \Phi_1 z_{t-12} - (\text{cross products}) + w_t$$

Write out the MA parts, the  $w_t$ .

$$w_t = e_t - \theta_1 e_{t-1} - \Theta_1 e_{t-12} - \Theta_2 e_{t-24} + (\text{cross products})$$

## auto.arima() for seasonal ts

auto.arima() will recognize that our data has season and fit a seasonal ARIMA model to our data by default. We will define the training data up to 1998 and use 1999 as the test data.

```
traindat <- window(chinookts, c(1990,10), c(1998,12))
testdat <- window(chinookts, c(1999,1), c(1999,12))
fit <- forecast::auto.arima(traindat)
fit
```

```
## Series: traindat
## ARIMA(1,0,0)(0,1,0)[12] with drift
##
## Coefficients:
##          ar1      drift
##      0.3676  -0.0320
## s.e.  0.1335   0.0127
##
## sigma^2 = 0.8053:  log likelihood = -107.37
## AIC=220.73   AICc=221.02   BIC=228.13
```

# Summary for seasonal models

Basic steps for identifying a seasonal model. **forecast** automates most of this.

- ▶ Check that you have specified your season correctly in your `ts` object.
- ▶ Plot your data. Look for trend, seasonality and random walks.



# Summary

- ▶ Use differencing to remove season and trend.
  - ▶ Season and no trend. Take a difference of lag = season
  - ▶ No seasonality but a trend. Try a first difference
  - ▶ Both. Do both types of differences
  - ▶ Neither. No differencing
  - ▶ Random walk. First difference
  - ▶ Parametric looking curve. Transform

# Summary

- ▶ Examine the ACF and PACF of the differenced data.
  - ▶ Look for patterns (spikes) at seasonal lags
- ▶ Estimate likely models and compare with model selection criteria (or cross-validation). Use `TRACE=TRUE`
- ▶ Do residual checks

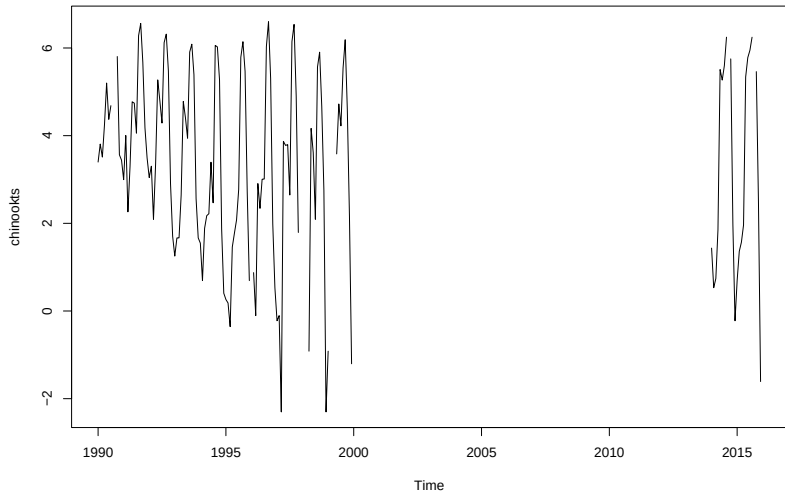
# Forecasting with a Seasonal model

Load the chinook salmon data

```
load("chinook.RData")  
chinookts <- ts(chinook$log.metric.tons, start=c(1990,1),  
               frequency=12)
```

## Plot seasonal data

```
plot(chinookts)
```



## Seasonal ARIMA model

Seasonally differenced data, e.g. chinook data January 1990 - chinook data January 1989.

$$z_t = x_t - x_{t+s} - m$$

Basic structure of a seasonal AR model

$$z_t = \text{AR}(p) + \text{AR}(\text{season}) + \text{AR}(p+\text{season})$$

$$\text{e.g. } z_t = \text{AR}(1) + \text{AR}(12) + \text{AR}(1+12)$$

Example AR(1) non-seasonal part + AR(1) seasonal part

$$z_t = \phi_1 z_{t-1} + \Phi_1 z_{t-12} - \phi_1 \Phi_1 z_{t-13}$$

# Notation

ARIMA  $(p,d,q)(ps,ds,qs)S$

ARIMA  $(1,0,0)(1,1,0)[12]$

## auto.arima() for seasonal ts

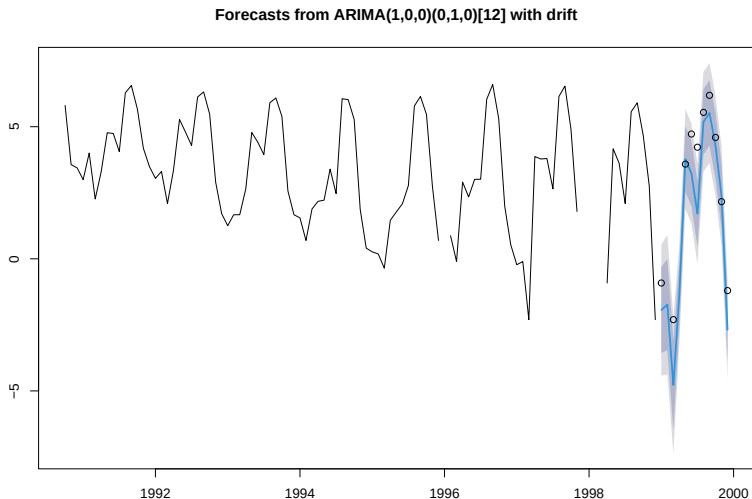
auto.arima() will recognize that our data has season and fit a seasonal ARIMA model to our data by default. We will define the training data up to 1998 and use 1999 as the test data.

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## Coefficients:
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##
## sigma^2 = 0.8053:  log likelihood = -107.37
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```

## Forecast using seasonal model

```
fr <- forecast::forecast(fit, h=12)  
plot(fr)  
points(testdat)
```





## Missing values

Missing values are ok when fitting a seasonal ARIMA model

